

DIU SMART TRANSPORT SYSTEM

This project is a **modern web-based Smart Bus Ticket Booking System named “DIU Smart Bus”** built using HTML, CSS, and JavaScript with localStorage-based CRUD functionality.

The system allows users to **register and log in using DIU student credentials**, after which they can access the dashboard to book bus seats. Users can select different bus routes based on morning and evening shifts, view available buses, and choose their preferred seats from an interactive seat layout.

The seat booking system visually displays a bus layout where users can select up to four seats. Booked seats are stored in the browser using localStorage, ensuring that already reserved seats are not available for other users. The system also dynamically calculates fare based on the number of selected seats and displays a real-time booking summary.

A **payment simulation module** is included where users can enter transaction details (bKash/Nagad) to confirm bookings. After successful booking, the system generates a **digital boarding pass (ticket)** showing passenger name, bus route, seat numbers, transaction ID, and total fare. Users can also download the ticket as a PDF using jsPDF and html2canvas.

The system includes full **CRUD operations**:

- **Create:** User registration and seat booking
- **Read:** Login, bus listing, and booking history
- **Update:** Profile update (name and password)
- **Delete:** Cancel booked tickets

Additionally, the application includes a booking history section where users can view and cancel their previous reservations. A responsive UI with animations, background slider, and modern glassmorphism design makes the system user-friendly and visually appealing.

Overall, this project demonstrates a complete front-end based smart transport management system with authentication, seat reservation, payment flow, ticket generation, and CRUD operations using browser storage.